

# CMSC3621 Data Structures and Algorithms Lab

## Lab Assignment 6

### Overview

This assignment is for the students to practice implementing the merge method in merge sort and partition method in quick sort.

### Details

In the attached .zip file, we provide a C++ project. You are to follow the comments of the functions to complete the code of `merge()` and `partition()` in the header file. When you finish the two functions, please run the code in `main()`, but do not change the code in `main.cpp`.

### Submission

- 1) `Sorting.h` //Comments are needed to illustrate how do you implement the code
- 2) A report (a PDF file), in the report you only need to provide screenshots and **descriptions** about how do you test your implementation and why your solution is correct according to the output

### Important requirements

- 1) Don't include any other files (that are not aforementioned).
- 2) Don't modify anything in the existing files / project, otherwise your code may not be able to run in the grader's environment.
- 3) You only can write your code at certain places which are indicated by the existing comments.
- 4) Any violation against the rules above will lead to extra penalty. If you missed the report, your submission will not be evaluated.
- 5) The following screenshot shows the correct output of the program. You must run you code in Visual Studio. As a result, the top-left corner of your command line window should display 'Microsoft Visual Studio ...' (as below). If you just reuse the screenshot provided below, you will get zero for this assignment.

```
Microsoft Visual Studio Debug Console

arr1:
-----
| 52 | 17 | 38 | 29 | 1 | 45 | 71 | 4 | 63 |
-----

arr1(Merge Sort):
-----
| 1 | 4 | 17 | 29 | 38 | 45 | 52 | 63 | 71 |
-----

arr2:
-----
| 26 | 89 | 12 | 6 | 23 | 90 | 2 | 56 | 43 |
-----

arr2(Quick Sort):
-----
| 2 | 6 | 12 | 23 | 26 | 43 | 56 | 89 | 90 |
-----

C:\Users\fzuo\source\repos\Sorting\x64\Debug\Sorting.exe (process 2848) exited with code 0.
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console when debugging stops.
```

---

## **Rubric**

1) There are two incomplete functions in total. Partial credits may be given in terms of to what degree you can complete them.

1.1) `merge()`: 40 pts

1.2) `partition()`: 45 pts

2) Report: 15 pts